

sage / BTP layering”). Each BTP segment SHALL be prepended with a BTP packet header and sent as the payload of a single GATT PDU. If a BTP SDU is split into more than one BTP segment, the BTP segments SHALL be sent in order of their position in the original BTP SDU, starting with the BTP segment at the buffer’s head.

At any point in time, only one BTP SDU may be transmitted in each direction over a BTP session. The transmission of BTP segments of any two BTP SDUs SHALL NOT overlap. If the application attempts to send one BTP SDU while transmission of another BTP SDU is in progress, the new BTP SDU SHALL be appended to a first-in, first-out queue. The next BTP SDU SHALL be dequeued from this queue and transmitted once transmission of the current BTP SDU completes, that is, once all BTP segments of the current BTP SDU have been transmitted and received by the peer via GATT.

The first BTP segment of a BTP SDU sent over a BTP session SHALL have its Beginning Segment header flag set to indicate the beginning of a new BTP SDU (see [Table 28, “BTP Packet PDU format”](#)). The presence of this flag SHALL indicate the further presence of a 16-bit unsigned integer field, the Message Length, that provides the receiver with the total length of the BTP SDU. The last BTP segment for a given BTP SDU SHALL have its Ending Segment flag set to indicate the end of the transmitted BTP SDU. A BTP packet that bears an unsegmented BTP SDU—that is, a BTP SDU small enough to fit into a single BTP segment—SHALL have both its Beginning Segment and Ending Segment flags set.

The size of a single BTP SDU sent via BTP is limited to 64KB, that is, the maximum size of the Message Length field in the BTP packet header. The number of segments used to send a buffer is unlimited and delimited by the Beginning Segment and Ending Segment bits in the BTP packet header. The upper layer imposes more stringent requirements over the maximum SDU size, such as [Section 4.4.4, “Message Size Requirements”](#).

The length of the data payload in each BTP segment whose Ending Segment bit is not set SHALL be equal to the session’s maximum BTP packet size minus the size of that packet’s header. If a packet’s Ending Segment bit is set, the length of its BTP segment data payload SHALL equal the size of the original BTP SDU minus the total size of all previously transmitted BTP segments of that BTP SDU. In this way, the length of a SDU’s last BTP segment is implied by its size.

Once a peer receives a complete set of BTP segments, it SHALL reassemble them in the order received, and verify that the reassembled BTP SDU’s total length matches that specified by the Beginning Segment’s Message Length value. If they match, the receiver SHALL pass the reassembled BTP SDU up to the next-higher-layer. If the reassembled buffer’s length does not match that specified by the sender, or if received BTP segment payload size would exceed the maximum BTP packet size, or receiver receives an Ending Segment without the presence of a previous Beginning Segment, or a Beginning Segment when another BTP SDU’s transmission is already in progress, the receiver SHALL close the BTP session and report an error to the application.

4.19.4.6. Sequence Numbers

All BTP packets SHALL be sent with sequence numbers, regardless of whether they contain SDU segments (for example, a packet acknowledgement with no attached segment payload). The purpose of sequence numbers is to facilitate the BTP receive window. A BTP sequence number SHALL be defined as an unsigned 8-bit integer value that monotonically increments by 1 with each packet sent by a given peer. A sequence number incremented past 255 SHALL wrap to zero.